

ADVANCED REAL ANALYSIS: I. MEASURE THEORY

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ABSTRACT

These are some notes I made while reading through Folland's book [Fol99]. Real Analysis is a very difficult subject, especially at the graduate level. As such, I figured I'd make them public so other learners have more resources to study.

Hopefully, graduate students and those studying for PhD qualifying examinations in Analysis will find them useful.

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Please also email me if you discover any errors.

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"The trick with education is to teach people in such a way that they don't realise that they are learning until it's too late." - Harold Eugene Edgerton

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MOTIVATION

Since antiquity, mathematicians have been obsessed with finding the areas of regions bounded by curves and volumes in three-dimensional Euclidean space.

This longing led to the development of calculus, spearheaded by Newton and Leibniz, and later Riemann, resulting in a drastic increase in our understanding of the Universe.

The Riemann integral is sufficient for calculating the areas (or volumes) of regions bounded by "sufficiently nice" functions, enabling it to be taught to millions of high school students globally, but fails in more extreme cases.

Recall that we say a bounded function on a closed interval is Riemann integrable if its lower Riemann integral equals its upper Riemann integral¹

It is not hard to find a function that is not Riemann integrable, and perhaps the most famous example is the Dirichlet function, defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

The non-integrability of this function is immediate from the density of $\mathbb{Q}$ in $\mathbb{R}$; any interval $[a, b]$ (where, of course, $a < b$), contains a real number and a rational number, so

$$\sup_{[a,b]} f = 1 \qquad \inf_{[a,b]} f = 0.$$

Consequently, for all partitions P of any closed interval $[c, d]$ in $\mathbb{R}$, $L(f, P, [c, d]) = 0$ and $U(f, P, [c, d]) = 1$. The upper and lower Riemann integrals of the Dirichlet function are, therefore,

$$U(f, [c, d]) = 1$$

and

$$L(f, [c, d]) = 0.$$

Citing page 9 of [Axl20], this example is disturbing because there are far fewer rational numbers than real numbers; recall that $\mathbb{Q}$ is countable, while $\mathbb{R}$ (and hence the set of irrationals) is uncountable²— intuitively, the integral of the Dirichlet function ought to be zero since the irrational contributions of "0" should drastically outweigh the rational contributions of "1".

¹The lower and upper Riemann integrals are defined by $L(f, [a, b]) = \sup_P L(f, P, [a, b])$, and $U(f, [a, b]) = \inf_P U(f, P, [a, b])$, for a bounded function $f : [a, b] \rightarrow \mathbb{R}$, respectively. Note the supremum is taken over all partitions P of the interval $[a, b]$.

²For an elegant proof of the uncountability of $\mathbb{R}$, see Cantor's Diagonalisation Argument.

Another issue is that the Dirichlet function (which I now restrict to $[0, 1]$ for simplicity) can be written as the pointwise limit of Riemann-integrable³ functions $f_k : [0, 1] \rightarrow \mathbb{R}$ defined by

$$f_k(x) = \begin{cases} 1, & x \in \{r_1, \dots, r_k\} \\ 0, & \text{otherwise,} \end{cases}$$

where $r_1, r_2, \dots$ is a sequence of each rational number⁴ (included only once) in $[0, 1]$. This shows the Riemann integral does not readily interchange with limits, as f is the pointwise limit of Riemann integrable functions, each being bounded by 1.

So is there another way? Well, to measure abstract volumes of sets, we would like to have a function μ that assigns a nonnegative real number, $\mu(E)$, to each subset $E \subset \mathbb{R}^n$ which satisfies countable additivity (which in turn implies finite additivity), meaning

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k)$$

for a (countably) disjoint sequence $\{E_k\}_{k=1}^{\infty}$ of subsets of $\mathbb{R}^n$.

Moreover, if E can be transformed into some other subset F through translations, rotations, and reflections, intuitively speaking, the size of E remains unchanged; the size of a ball's mirror image is the same as its own size. Hence, we would like $\mu(E) = \mu(F)$.

Last, it would make sense for $\mu(Q) = 1$, where Q denotes the unit cube in $\mathbb{R}^n$ —see page 19 of [Fol99].

Unfortunately, these definitions are inconsistent with one another (see page 20 of Folland). For an even more depressing example, one may look through 2.22 in [Axl20] to show that $\mathbb{R}$ (and hence $\mathbb{R}^n$) contains such pathological subsets that it is unreasonable for us to demand that μ be defined on *all* subsets of $\mathbb{R}^n$. This brings us to a discussion of σ -algebras.

1. SIGMA ALGEBRAS

Definition: σ -Algebras

Given a set X , we say that a set *of subsets* of X , $\mathcal{M}$, is a σ -algebra on X if we have the following:

- (1) $\emptyset \in \mathcal{M}$
- (2) If $E \in \mathcal{M}$, then $E^c \in \mathcal{M}$
- (3) If $\{E_k\}_{k=1}^{\infty}$ is a sequence of elements in $\mathcal{M}$, then the union $\bigcup_{k=1}^{\infty} E_k \in \mathcal{M}$.

Here is an example of a σ -algebra:

³The set of discontinuities of f_k for each $k \in \mathbb{N}$ is finite.

⁴We may do this as $\mathbb{Q}$ is countable; see Theorem 1.5.6(i) in [Abb15].

Theorem 1.

Let X be an uncountable set. Then the collection

$$\mathcal{M} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra on X .

Proof: $\emptyset \in \mathcal{M}$ and in particular, $\emptyset$ is countable, so that $\emptyset \in \mathcal{M}$.

Let $E \in \mathcal{M}$ be given. Now, if E is countable, $(E^c)^c = E$ is countable, so that $E^c \in \mathcal{M}$. If E^c is countable, that immediately satisfies our requirement.

To verify the last condition for a σ -algebra, let $\{E_n\}_{n=1}^\infty$ be a family of elements in $\mathcal{M}$. We proceed by cases:

Case 1: All the E_n 's are countable

Since the countable union of countable sets is countable⁵, then

$$\bigcup_{n=1}^\infty E_n \in \mathcal{M}$$

by definition.

Case 2: At least one E_n is only co-countable.

Then, since

$$E_n \subset \bigcup_{k=1}^\infty E_k,$$

taking complements yields

$$\left(\bigcup_{k=1}^\infty E_k \right)^c \subset E_n^c$$

so by assumption on E_n , $(\bigcup_{k=1}^\infty E_k)^c$ is a subset of a countable set, and is therefore countable. That is,

$$\bigcup_{k=1}^\infty E_k$$

is co-countable (and therefore in $\mathcal{M}$).

Cases 1 and 2 highlight that $\bigcup_{k=1}^\infty E_k$ is always contained in $\mathcal{M}$, so that by definition, $\mathcal{M}$ is a σ -algebra on X . ■

⁵See Theorem 1.5.8 (ii) in [Abb15]

2. MEASURES

One of the biggest problems in Mathematics is finding a way to quantify size. Abstract *distances* can be understood in terms of metrics, which were then used to develop the theory of functions on metric spaces; see Chapter 19 of [Tao16] for a quick review.

We now turn to the issue of finding an abstract notion of *size*, or rather, *volume*.

Just as we defined a *metric* $d : X \times X \rightarrow [0, \infty]$ on a set X to satisfy certain properties⁶ that are in accordance with the familiar notion of distance in Euclidean space, we wish to define a generalised "*volume*" function $\mu : \mathcal{M} \rightarrow [0, \infty]$ ⁷ over a σ -algebra $\mathcal{M}$ which assigns an abstract volume $\mu(E)$ to a set $E \in \mathcal{M}$; recall that the σ -algebra is a family of sets.

Now, how do we generalise the size of a set? That is to say, given a σ -algebra $\mathcal{M}$, what properties should a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ have?

For anything to be endowed with the responsibility of giving a generalised volume to a set, it must surely assign the value "0" to the empty set $\emptyset$, as it has no elements. Intuitively, if you imagine an empty water tank, then the amount of water inside, according to any reasonable scale of measurement, must be zero.

We therefore impose our first condition on μ : $\mu(\emptyset) = 0$.

What are some other desirable properties? Consider a sequence $\{E_k\}_{k=1}^{\infty}$ of disjoint sets in the σ -algebra. We ought to have the relation

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k).$$

To see why this must hold intuitively, consider cutting a cake into however many parts you like. Then certainly the sum of " μ " of all the slices would equal " μ " of the entire cake, which is equivalent to the *union* of all the pieces!

⁶Symmetry, $d(x, y) = 0$ if and only if $x = y$, and the Triangle Inequality.

⁷We relegate the image of μ to the nonnegative real line as it doesn't make sense for an object to possess negative volume or if you wish, size.

We therefore require μ to satisfy the two conditions discussed above, and call such functions a *measure*:

Definition: Measures

Let X a set equipped with a σ -algebra $\mathcal{M}$. A *measure* on $\mathcal{M}^a$ is a function $\mu : \mathcal{M} \rightarrow [0, \infty]$ such that

- (1) $\mu(\emptyset) = 0$
- (2) If $\{E_k\}_{k=1}^\infty$ is a sequence of disjoint sets in $\mathcal{M}$, then

$$\mu\left(\bigcup_{k=1}^\infty E_k\right) = \sum_{k=1}^\infty \mu(E_k).$$

^aIf $\mathcal{M}$ is clear from the context, we may simply say "on X ."

One remark: " $\mu(\bigcup_{k=1}^\infty E_k)$ " in condition (2) is well-defined because σ -algebras are closed under countable union.

(2) in the definition above immediately implies finite additivity, that is,

$$\mu\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n \mu(E_k)$$

for a finite collection of disjoint sets $E_1, \dots, E_n \in \mathcal{M}$, because we may take $\{F_k\}_{k=1}^\infty$ to be

$$F_k = \begin{cases} E_k & , 0 \leq k \leq n, \\ \emptyset & , \text{otherwise.} \end{cases}$$

For X a set and $\mathcal{M} \subset \mathcal{P}(X)$ a σ -algebra, we call the pair $(X, \mathcal{M})$ a *measurable space*. Sets contained in $\mathcal{M}$ are referred to as *measurable*. If μ is a measure on $(X, \mathcal{M})$, then the triple $(X, \mathcal{M}, \mu)$ is called a *measure space*.

Functions $\mu : \mathcal{M} \rightarrow [0, \infty]$ that satisfy (1) and *finite* additivity (not necessarily (2)) are called *finite* measures.

Here is an example of a measure, as outlined in this theorem.

Theorem 1.

Let X be an uncountable set, and let $\mathcal{M}$ be the σ -algebra of countable or co-countable sets.⁸

Define the function $\mu : \mathcal{M} \rightarrow [0, \infty]$ by

$$\mu(E) = \begin{cases} 0, & E \text{ countable} \\ 1, & E \text{ co-countable,} \end{cases}$$

where $E \subset X$. Then μ constitutes a measure on $(X, \mathcal{M})$.

⁸This was discussed in Section 1, Theorem 1.

Proof: The empty set is countable, so $\mu(\emptyset) = 0$. To prove countable additivity, consider first a disjoint sequence $\{E_k\}_{k=1}^{\infty}$ of countable sets in X . The countable union of countable sets is countable, so that

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 0 = \sum_{k=1}^{\infty} \mu(E_k).$$

Now, if we had some $j \in \mathbb{N}$ such that E_j is co-countable, the union $\bigcup_{k=1}^{\infty} E_k$ is also co-countable (see Case 2 in Section 1, Theorem 1). Moreover, disjointness implies that $E_k \subset E_j^c$ for all $k \neq j$, i.e., the E_k 's are subsets of a countable set for $k \neq j$, and are therefore countable. By definition of μ ,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 1, \quad \sum_{k=1}^{\infty} \mu(E_k) = \mu(E_j) = 1,$$

so in general, we have shown, for a disjoint sequence $\{E_k\}_{k=1}^{\infty}$,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{k=1}^{\infty} \mu(E_k),$$

that is, μ possesses countable additivity. ■

The Dirac Measure.

Let X be a nonempty set, and consider the σ -algebra $\mathcal{M} = \mathcal{P}(X)$ (the power set). Choose an $x_0 \in X$ and define the function $f : X \rightarrow [0, \infty]$ by

$$f(x) = \begin{cases} 1, & x = x_0 \\ 0, & x \neq x_0. \end{cases}$$

Then the map $\mu : \mathcal{M} \rightarrow [0, \infty]$ defined by $\mu(E) = \sum_{x \in E} f(x)$ is a measure on $\mathcal{M}$.

Proof:

$$\mu(\emptyset) = \sum_{x \in \emptyset} f(x) = 0.$$

For countable additivity, suppose we are given an arbitrary collection $\{E_k\}_{k=1}^{\infty}$ of disjoint sets contained in $\mathcal{M}$. If *none* of these sets contain x_0 , then

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 0 = \sum_{k=1}^{\infty} \mu(E_k),$$

since $x_0 \notin \bigcup_{k=1}^{\infty} E_k$.

If one of these sets contains x_0 , then by disjointness none of the others contain x_0 . This implies

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = 1 = \sum_{k=1}^{\infty} \mu(E_k).$$

This proves countable additivity. ■

The Dirac Measure, named after the Nobel Prize-winning physicist Paul Dirac⁹, can be used to rigorously define the Dirac Delta Function, which has numerous applications in Physics.

In general, a function $f : X \rightarrow [0, \infty]$ determines a measure μ by the sum

$$\mu(E) = \sum_{x \in E} f(x).$$

Indeed, for $E = \emptyset$, we have $\mu(\emptyset) = 0$, which can be easily deduced from the equation above. Countable additivity holds, because if $\{E_k\}_{k=1}^{\infty}$ is a collection of disjoint sets in $\mathcal{M}$,

$$\mu\left(\bigcup_{k=1}^{\infty} E_k\right) = \sum_{x \in \bigcup_{k=1}^{\infty} E_k} f(x) = \sum_{k=1}^{\infty} \sum_{x \in E_k} f(x) = \sum_{k=1}^{\infty} \mu(E_k),$$

where the second equality holds by disjointness.

The next result shows that if $\mu_1, \dots, \mu_n$ are measures on a σ -algebra $\mathcal{M}$, and $a_1, \dots, a_n$ are nonnegative constants, then $\mu := a_1\mu_1 + \dots + a_n\mu_n$ is a measure on $\mathcal{M}$.

The Linear Combination of Measures is Also a Measure.

Suppose X is a set with a σ -algebra $\mathcal{M}$, and that $\mu_1, \dots, \mu_n$ are measures on $\mathcal{M}$. Then, for any nonnegative $a_1, \dots, a_n$, the function $\mu : \mathcal{M} \rightarrow [0, \infty]$ defined by

$$\mu := \sum_{k=1}^n a_k \mu_k$$

is a measure on $\mathcal{M}$.

Proof: Observe

$$\mu(\emptyset) = \sum_{k=1}^n a_k \mu_k(\emptyset) = 0.$$

This verifies the first property. For the second property, suppose $\{E_n\}_{n=1}^{\infty}$ is a sequence of disjoint sets in $\mathcal{M}$. Then

⁹Dirac predicted the existence of the positron, which was later confirmed through experiment. He is also credited with laying the groundwork in Quantum Field Theory.

$$\begin{aligned}
\mu\left(\bigcup_{n=1}^{\infty} E_n\right) &= \sum_{k=1}^n a_k \mu_k\left(\bigcup_{n=1}^{\infty} E_n\right) \\
&= \sum_{k=1}^n a_k \sum_{n=1}^{\infty} \mu_k(E_n) \\
&= \sum_{k=1}^n \sum_{n=1}^{\infty} a_k \mu_k(E_n).
\end{aligned}$$

All terms in the double sum above are nonnegative by assumption, so we may interchange the order of summation:

$$\begin{aligned}
\mu\left(\bigcup_{n=1}^{\infty} E_n\right) &= \sum_{n=1}^{\infty} \sum_{k=1}^n a_k \mu_k(E_n) \\
&= \sum_{n=1}^{\infty} \mu(E_n),
\end{aligned}$$

by our definition of μ . ■

Measures have some useful properties, as outlined in this theorem.

Theorem 2 (Folland 1.8).

Let $(X, \mathcal{M}, \mu)$ be a measure space. Then,

- (1) (*Monotonicity*)
If $E, F \in \mathcal{M}$ and $E \subset F$, then $\mu(E) \leq \mu(F)$.
- (2) (*Subadditivity*)
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$, then $\mu(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} \mu(E_k)$.
- (3) (*Continuity from below*)
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \dots$, then $\mu(\bigcup_{k=1}^{\infty} E_k) = \lim_{k \rightarrow \infty} \mu(E_k)$.
- (4) (*Continuity from above*)
If $\{E_k\}_{k=1}^{\infty} \subset \mathcal{M}$ and $E_1 \supset E_2 \supset \dots$, and $\mu(E_1) < \infty$, then $\mu(\bigcap_{k=1}^{\infty} E_k) = \lim_{k \rightarrow \infty} \mu(E_k)$.

The proof of this result can be found on page 27 of [Fol99].

Before we proceed, a discussion on finite measures.

Recall that a map $\mu : \mathcal{M} \rightarrow [0, \infty]$ that satisfies $\mu(\emptyset) = 0$ and

$$\mu\left(\bigcup_{j=1}^n E_j\right) = \sum_{j=1}^n \mu(E_j)$$

for any finite collection of disjoint sets $\{E_j\}_{j=1}^n$ in the σ -algebra $\mathcal{M}$ (but not necessarily countable additivity) is referred to as a *finite* measure; it is a measure up to finite unions.

Let $\{E_j\}_{j=1}^\infty \in \mathcal{M}$ be a sequence of disjoint sets. We observe the inclusion

$$E_1 \subset E_1 \cup E_2 \subset E_1 \cup E_2 \cup E_3 \subset \dots$$

If we define the collection $\{F_k\}_{k=1}^\infty$ by

$$F_k = \bigcup_{j=1}^k E_j,$$

we notice that $F_1 \subset F_2 \subset \dots$, and $\bigcup_{k=1}^\infty F_k = \bigcup_{j=1}^\infty E_j$.

As μ has finite additivity,

$$\mu(F_k) = \sum_{j=1}^k \mu(E_j).$$

Now look at (3) in Theorem 2. Our discussion above implies that μ is a measure if it is continuous from below, because

$$\mu\left(\bigcup_{k=1}^\infty F_k\right) = \mu\left(\bigcup_{j=1}^\infty E_j\right) = \lim_{k \rightarrow \infty} \mu(F_k) = \lim_{k \rightarrow \infty} \sum_{j=1}^k \mu(E_j) = \sum_{j=1}^\infty \mu(E_j)$$

and μ therefore admits countable additivity.

3. ANALOGY TO FINITE-DIMENSIONAL VECTOR SPACES

The measure of two sets is analogous to the dimension of the sum of two finite-dimensional vector subspaces; first recall the familiar theorem from linear algebra¹⁰

Dimension of Two Subspaces.

Let V_1 and V_2 be subspaces of a finite-dimensional vector space. Then

$$\dim(V_1 + V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2).$$

If we view the dimension as the size of the vector space, then the formula above is reminiscent of how (long ago) we were taught to find the area of two intersecting shapes: You first find the sum of their two individual areas, and then subtract the area of the region where they overlap.

A similar result holds for measures:

¹⁰For a proof, see 2.37 in [Axl24]

Theorem 3.

If $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$, then

$$\mu(E \cup F) = \mu(E) + \mu(F) - \mu(E \cap F),$$

provided $\mu(E \cap F) < \infty$.

To prove this result, we need a lemma:

Lemma 1. Suppose $(X, \mathcal{M}, \mu)$ is a measure space and $E, F \in \mathcal{M}$ are such that $E \subset F$. Then, if $\mu(E) < \infty$, $\mu(F \setminus E) = \mu(F) - \mu(E)$.

Proof: As $F = E \cup (F \setminus E)$ and this union is disjoint, $\mu(F) = \mu(E) + \mu(F \setminus E)$, so since $\mu(E) < \infty$, we may subtract $\mu(E)$ from both sides to obtain

$$\mu(F \setminus E) = \mu(F) - \mu(E),$$

as desired. $\square$

Proof of Theorem 3: We may write

$$E \cup F = (E \setminus (E \cap F)) \cup (F \setminus (E \cap F)) \cup (E \cap F).$$

Since the union above is disjoint, finite additivity of the measure implies

$$\mu(E \cup F) = \mu(E \setminus (E \cap F)) + \mu(F \setminus (E \cap F)) + \mu(E \cap F).$$

The intersection $E \cap F$ is a subset of *both* E and F , so invoking Lemma 1,

$$\begin{aligned} \mu(E \cup F) &= (\mu(E) - \mu(E \cap F)) + (\mu(F) - \mu(E \cap F)) + \mu(E \cap F) \\ &= \mu(E) + \mu(F) - \mu(E \cap F) \end{aligned}$$

which is the desired result. $\blacksquare$

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